Software Requirements Specification (SRS)

Graduate Project Research & Recommendation System (GPRR)

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Graduate Project Redundancy Reduction (GPRR) System, a platform designed to eliminate duplicate and redundant graduation projects through AI-powered similarity detection, while facilitating team collaboration and faculty oversight in academic institutions.

1.2 Scope

The GPRR system's primary goal is to reduce redundancy in graduation projects by detecting similar or duplicate work before and during project development. The system enables students to form teams, propose projects within specific domains and technologies, and automatically verifies project originality against existing and historical projects using vector embeddings and semantic similarity analysis. The system supports three user roles: Students, Professors, and Administrators.

1.3 Definitions and Acronyms

- **GPRR**: Graduate Project Redundancy Reduction
- **ERD**: Entity Relationship Diagram
- **GPA**: Grade Point Average
- **SRS**: Software Requirements Specification
- **API**: Application Programming Interface
- **Redundancy**: The occurrence of multiple similar or duplicate graduation projects
- **Vector Embedding**: Mathematical representation of project content for similarity comparison
- **Originality Score**: Percentage indicating how unique a project is compared to existing projects

1.4 System Overview

The GPRR system addresses the critical problem of redundant graduation projects in academic institutions. By leveraging AI-powered vector embeddings and semantic analysis, the system proactively identifies similar projects, prevents duplicate work, and guides students toward original research areas. The platform manages the entire project lifecycle—from initial proposal to final submission—with continuous redundancy checking at each stage.

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2. System Features and Requirements

2.1 User Management

2.1.1 User Registration and Authentication

Priority: High

Description: The system shall support secure user registration and authentication for all user types.

Functional Requirements:

- FR-UM-001: System shall allow new users to register with email, password, first name, and last name
- FR-UM-002: System shall hash and securely store user passwords
- FR-UM-003: System shall validate email format and uniqueness during registration
- FR-UM-004: System shall support login using email and password credentials
- FR-UM-005: System shall track user account creation timestamp
- FR-UM-006: System shall support account activation/deactivation via Is_Active flag
- FR-UM-007: System shall allow users to upload and update profile pictures

2.1.2 Role-Based Access Control

Priority: High

Description: The system shall implement role-based access control for Students, Professors, and Administrators.

Functional Requirements:

- FR-UM-008: System shall assign exactly one role to each user (Student, Professor, or Admin)
- FR-UM-009: System shall restrict feature access based on assigned user role
- FR-UM-010: Each Student profile shall store graduation year, GPA, and role information
- FR-UM-011: Each Professor profile shall store department affiliation and specialization
- FR-UM-012: Each Admin profile shall be linked to a user account

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2.2 Department Management

2.2.1 Department Operations

Priority: Medium

Description: The system shall manage academic departments and their associations with users.

Functional Requirements:

- FR-DM-001: System shall maintain departments with unique IDs, names, and codes
- FR-DM-002: System shall allow Students to enroll in departments
- FR-DM-003: System shall allow Professors to work in departments
- FR-DM-004: System shall support multiple students per department
- FR-DM-005: System shall support multiple professors per department

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2.3 Skills Management

2.3.1 Student Skills Tracking

Priority: Medium
Description: The system shall track and manage student skills for profile purposes and filtering.
Functional Requirements:

- FR-SM-001: System shall allow students to add multiple skills to their profile
- FR-SM-002: Each skill shall have a unique identifier and name
- FR-SM-003: System shall associate skills with specific students
- FR-SM-004: System shall enable searching/filtering students by skills
- FR-SM-005: Skills shall be displayed on student profiles
- FR-SM-006: Students shall be able to add or remove skills from their profile

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2.4 Team Management

2.4.1 Team Creation and Configuration

Priority: High
Description: The system shall enable team formation with configurable parameters.
Functional Requirements:

- FR-TM-001: System shall allow creation of teams with unique names
- FR-TM-002: Each team shall have a unique join code for member recruitment
- FR-TM-003: System shall enforce maximum team size limits
- FR-TM-004: System shall track team creation timestamp
- FR-TM-005: System shall associate teams with supervising professors

2.4.2 Team Membership

Priority: High
Description: The system shall manage team membership and roles.
Functional Requirements:

- FR-TM-006: System shall allow students to join teams via join codes
- FR-TM-007: System shall track when each member joined the team
- FR-TM-008: System shall assign roles to team members (e.g., leader, member)
- FR-TM-009: System shall prevent team size from exceeding maximum limit
- FR-TM-010: System shall maintain team member status and participation record

2.4.3 Join Requests

Priority: Medium

Description: The system shall manage team join requests with approval workflow.

Functional Requirements:

- FR-TM-011: Students shall be able to send join requests to teams
- FR-TM-012: System shall track request status (pending, approved, rejected)
- FR-TM-013: System shall timestamp all join requests
- FR-TM-014: Team members with appropriate permissions shall approve/reject requests
- FR-TM-015: System shall notify students of request status changes

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2.5 Project Management

2.5.1 Project Creation and Details

Priority: High

Description: The system shall support comprehensive definition and tracking of graduation projects, with each team developing exactly one graduation project.

Functional Requirements:

- FR-PM-001: System shall allow teams to create ONE graduation project with title and description
- FR-PM-002: System shall enforce that each team can have only one active project
- FR-PM-003: Each project shall include an abstract summarizing the work
- FR-PM-004: System shall categorize projects by domain
- FR-PM-005: System shall track project status (draft, in-progress, submitted, completed)
- FR-PM-006: Projects must receive professor approval before status can change from "draft" to "in-progress"
- FR-PM-007: System shall record project creation and submission timestamps
- FR-PM-008: System shall calculate and store originality scores for each project
- FR-PM-009: System shall prevent team from creating additional projects once one exists

2.5.2 Project Domains and Technologies

Priority: High

Description: The system shall categorize projects by domain and track technologies used.

Functional Requirements:

- FR-PM-007: System shall maintain a catalog of domains with names and descriptions
- FR-PM-008: Each project shall be assigned to one primary domain
- FR-PM-009: System shall maintain a catalog of technologies with names and categories
- FR-PM-010: System shall associate multiple technologies with each project
- FR-PM-011: System shall enable filtering projects by domain and technology

2.5.3 Chat Room File Attachments

Priority: Medium

Description: The system shall support file sharing within team chat rooms for collaboration.

Functional Requirements:

- FR-PM-012: System shall allow team members to upload files in chat rooms
- FR-PM-013: System shall store file metadata (name, type, size, URL)
- FR-PM-014: System shall timestamp file uploads
- FR-PM-015: System shall support common file formats (PDF, DOCX, images, code files, compressed files)
- FR-PM-016: System shall enforce file size limits per attachment (configurable, default 25MB)
- FR-PM-017: System shall display attachments inline within chat messages where applicable
- FR-PM-018: System shall allow team members to download shared files
- FR-PM-019: Attachments may include meeting notes, reference materials, code snippets, images, or any team collaboration files

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2.6 Originality Detection & Redundancy Reduction

2.6.1 Similarity Analysis

Priority: Critical

Description: The system's core function is to detect and prevent redundant graduation projects through AI-powered similarity analysis.

Functional Requirements:

- FR-OD-001: System shall automatically generate originality reports when projects are created or updated
- FR-OD-002: System shall calculate an overall originality score (0-100%) where higher scores indicate more original work
- FR-OD-003: System shall provide a detailed summary of similarity findings with specific areas of overlap
- FR-OD-004: System shall identify all similar existing projects with similarity percentages above a configurable threshold
- FR-OD-005: System shall explain match reasons for each similar project (e.g., similar domain, shared technologies, comparable methodology)
- FR-OD-006: System shall timestamp all report generations for audit purposes
- FR-OD-007: System shall flag projects with originality scores below 70% for review
- FR-OD-008: System shall prevent project submission if originality score is below minimum threshold (configurable, default 60%)
- FR-OD-009: System shall compare projects against all historical projects in the database
- FR-OD-010: System shall alert students and advisors when high similarity is detected

2.6.2 Vector Embeddings for Semantic Analysis

Priority: Critical

Description: The system shall use advanced vector embeddings to detect semantic similarity beyond simple keyword matching.

Functional Requirements:

- FR-OD-011: System shall generate vector embeddings for all project content (title, abstract, description)
- FR-OD-012: System shall store the AI model name and version used for embedding generation
- FR-OD-013: System shall store high-dimensional vector data for precise similarity comparisons
- FR-OD-014: System shall timestamp embedding generation for tracking updates
- FR-OD-015: System shall automatically regenerate embeddings when project content changes significantly
- FR-OD-016: System shall use cosine similarity or equivalent metrics to compare vector embeddings
- FR-OD-017: System shall detect semantic similarity even when exact wording differs
- FR-OD-018: System shall maintain embedding consistency across all projects for fair comparison

2.6.3 Redundancy Prevention Workflow

Priority: High

Description: The system shall implement a proactive workflow to prevent redundant projects before significant work begins.

Functional Requirements:

- FR-OD-019: System shall perform preliminary similarity check when project idea is first submitted
- FR-OD-020: System shall provide early warning to students if similar projects exist
- FR-OD-021: System shall suggest alternative directions or modifications to increase originality
- FR-OD-022: System shall require professor approval for projects with originality scores between 60-70%
- FR-OD-023: System shall maintain a reference database of all current and past projects
- FR-OD-024: System shall archive completed projects for future comparison

2.7 Communication System

2.7.1 Notifications

Priority: High

Description: The system shall provide a notification system for important events.

Functional Requirements:

- FR-CS-001: System shall send notifications for team invitations
- FR-CS-002: System shall send notifications for join request status changes
- FR-CS-003: System shall send notifications for new feedback
- FR-CS-004: System shall send notifications for project status changes
- FR-CS-005: Each notification shall include type, title, message, and timestamp
- FR-CS-006: System shall track read/unread status for notifications
- FR-CS-007: Users shall be able to view notification history

2.7.2 Chat System with File Sharing

Priority: Medium

Description: The system shall provide real-time chat functionality with file sharing for teams.

Functional Requirements:

- FR-CS-008: Each team shall have an associated chat room created automatically upon team formation
- FR-CS-009: System shall track chat room creation timestamp
- FR-CS-010: Team members shall be able to send text messages in their team's chat room
- FR-CS-011: Team members shall be able to attach files to chat messages
- FR-CS-012: System shall store message content and sender information
- FR-CS-013: System shall timestamp all messages
- FR-CS-014: System shall display messages in chronological order
- FR-CS-015: System shall support real-time message delivery
- FR-CS-016: Attached files shall be accessible to all team members
- FR-CS-017: System shall show file preview/thumbnails where applicable
- FR-CS-018: Chat room files are for general team communication and collaboration, not limited to project documentation

2.8 Feedback System

2.8.1 Feedback Management

Priority: High

Description: The system shall enable professors to provide feedback on projects.

Functional Requirements:

- FR-FS-001: Professors shall be able to provide feedback on projects
- FR-FS-002: Each feedback entry shall include content and creation timestamp
- FR-FS-003: System shall categorize feedback by type (technical, presentation, documentation, etc.)
- FR-FS-004: System shall associate feedback with specific projects
- FR-FS-005: Students shall be able to view all feedback received on their projects
- FR-FS-006: System shall track which professor provided each feedback item